Telemetry-based Analysis of Single-Event-Induced Failures in Unmanned Aerial Vehicles

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***Abstract*—Telemetry-based data collection and analysis is used to characterize single-event effects for commercially-available drones for terrestrial-neutron spectra. Drones were irradiated and tested under the ChipIR beamline at the ISIS facility. Multiple parameters that are vital to a drone's flight were monitored using telemetry data collected during beam exposure. Telemetry data for baseline operations (without the neutron beam) were also obtained under the same operating conditions for comparison. In total, out of 28 runs, 26 runs showed anomalies in telemetry data that deviated from the baseline data and revealed time gaps between telemetry and visual anomalies. Telemetry data was then analyzed to identify which parameters most commonly showed anomalies first during all runs and for different failure characteristics. Possible reasons for observed telemetry data patterns are hypothesized. FIT rates for different failure mechanisms are also calculated and show very high vulnerability.**

***Index Terms*— Single event effect, single event upset, unmanned aerial vehicle**

I. Introduction

DRONES or Unmanned Aerial Vehicles (UAVs) are airborne systems that are controlled remotely or are autonomous with onboard electronic systems. Drones are projected to play very important roles in commercial and Mil-Aero applications in the future [1]. The current usage of drones by the commercial sector involves the transportation of goods without human intervention. Amazon has begun using drones to deliver small packages to homes in a project named "Prime Air". Many other commercial companies with an online marketplace, such as Walmart, have followed Amazon's lead by using drones in their own delivery system. Current usage for military applications has far outpaced those for commercial applications. The USA has a large drone fleet for remote sensing and targeted surgical strikes. In the Russia-Ukraine War, both Russian and Ukraine military forces are using drones for targeting enemy positions and equipment. Ukraine has also deployed drones effectively for defensive purposes. It is evident that reliability for these drones is of utmost importance for both commercial as well as Mil-Aero applications. Any deviation from expected behavior will have severe consequences for all these applications.

These drones are designed and operated in the terrestrial environment where they experience single-event effects (SEE) due to terrestrial neutrons, thermal neutrons, and alpha particles. Since drones are mostly assembled with commercial-off-the-shelf (COTS) parts with functionality as the primary objective, a lack of attention to SEE may result in the loss of drone functionality. Additionally, most drones use autonomous onboard control systems to manage routine flight functions. These control systems use sensory data such as altitude, position, velocity, yaw, pitch, roll, temperature, airspeed, etc., to calculate the correct parameters to keep the drone afloat and on the proper trajectory to the destination. SEE may interfere in drone operations with catastrophic consequences, such as a crash, premature ordnance drops, or veering off the intended path. Though the functionality of these drones is thoroughly verified under normal operating conditions, the effects of single-event upsets (SEU) on drones have not been studied in detail previously. This work presents the effects of terrestrial neutron exposures for autonomous drones to identify different failure modes and failure rates due to SEE based on telemetry data. Telemetry data from drone sensors, along with audio and video feeds, during radiation exposures are used to identify failure initiation and failure characteristics. Results show very high FIT rates for drones used in the test.

II. Background

Drones, particularly multi-copters, are inherently unstable due to the drone's reliance on rotors to constantly stabilize and compensate for multiple factors, such as gravity, wind, centrifugal forces, load oscillation, and other forces impacting the flight of the drone. This means an active onboard stabilization system is necessary for proper operation [2]. Without active stabilization, the drone will crash. Electronic systems for most multi-copter drones are designed with four major circuit blocks, with each block responsible for specific

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functions. These blocks are (i) a flight controller, (ii) a motor controller, (iii) a communications module, and (iv) a sensor module.

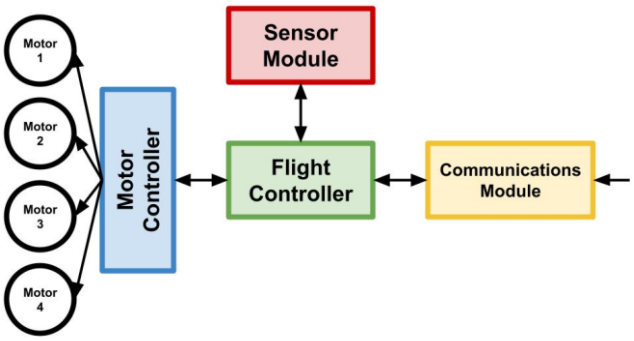


Fig. 1. Four main modules of a typical drone.

Fig. 1 shows the block diagram for communications links between these modules for a typical drone. The flight controller is the brain of the system responsible for keeping the drone afloat based on sensor data and/or commands received from the pilot. The flight controller is also responsible for the active stabilization of the drone. Firmware storage for all circuit blocks is considered part of the flight controller module. The motor controller determines the speed of each motor based on a desired flight path, stabilization control, and sensor signals. The communications module is responsible for all communications between the base station and the drone (including telemetry data). The sensor module provides data for the flight controller, such as acceleration in all 3 dimensions, yaw, pitch, roll, elevation, temperature, wind speed (on some drones), etc. The sensor module also logs internal data, such as individual rotor speed, the direction of flight, etc. All four blocks mentioned above need to interact seamlessly and faultlessly for a successful flight of the drone.

Flight controllers are usually microcontrollers with firmware for all necessary functions. The motor controller may also contain a micro-controller with its own firmware. The communications module often use Wi-Fi or other protocols, typically the most recent one available at the time the drone was designed. Previous work on micro-controllers has shown significant vulnerability to single-event effects [3-6] stemming from the primary requirement of low cost for micro-controllers [7]. Though SEE vulnerability of micro-controllers has been investigated previously, it was done either as a stand-alone component or as part of other electronic systems [8-10]. Because drone operation requires close interaction between all the above-mentioned circuit blocks, results from individual components are not sufficient to evaluate the overall system-level effects [11,12]. Computer simulation of the whole drone system has been carried out and system recovery techniques suggested through replication of the state of the micro-controllers [13,14]. The reliability of communications between the receiver and transmitter in drones has also been researched using statistical modeling and fault injections [15,16]. But these simulation models need to be verified through actual experiments by exposing drones to a beamline. This work has evaluated SE vulnerability of a drone through analysis of telemetry data when exposed to terrestrial neutrons to identify failure modes, elucidate failure mechanisms, and develop mitigation techniques.

III. Experiments

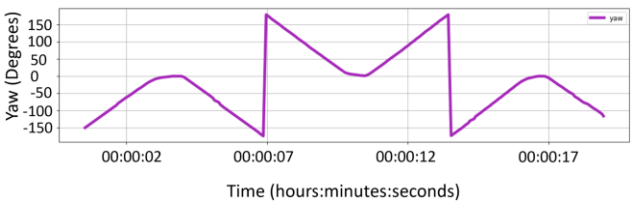
For this work, Tello EDU drones were selected because of the ease of telemetry data collection [17]. Telemetry data is necessary to identify when failure started and what caused the final operational failure. Without extensive telemetry data, typical audio and video signals available during operation may not be sufficient for failure classification. This drone exposes an application programming interface (API) allowing control of the drone as well as for telemetry data collection. [18] All the code necessary for this project has been developed in Python.

Atmospheric neutron irradiation experiments were carried out at the ChipIR beamline at the ISIS Facility, UK. ChipIR is a beamline dedicated to the irradiation of microelectronics that has a fast-neutron beam that mimics that of atmospheric and terrestrial neutron spectra but with a neutron flux of 5.6 x10^6 neutrons/cm^2/sec above 10 MeV [19, 20]. 3 Tello EDU drones were consecutively exposed to the beam at 3 different locations from the beam port (most forward, middle, and most back positions). The first drone position was 14 m away from the target, the last drone was 18.3 m away from the target. Though the beam flux will decrease with increasing distance from the target, there were no significant differences found in testing based on drone position. During exposure, the drone was programmed to fly within a 10 cm x 10 cm x 10 cm volume area with the beam collimation set to ensure the drone's continuous irradiation during the extent of its flight within this volume. During irradiations, drone was programmed for a clockwise rotation followed by counterclockwise rotation while maintaining elevation. The drones were fitted with safeguards, such as plastic cages or propeller guards, to minimize damage during crashes. Each drone was power cycled after each test run. A test run was considered complete if the drone stayed afloat and the battery ran out (~15 minutes) or an operational failure was observed. During exposure, the drone was connected to a laptop using 802.11n Wi-Fi protocol. Python code on the laptop controlled the drone as well as collected telemetry data. Telemetry data from the drone was collected at 100 Hz for position, velocity, and acceleration in all 3 dimensions, yaw, pitch, roll, elevation and height, temperature of the drone, battery percentage, and the time duration of the flight.

In addition to telemetry data, a video camera in the beam room recorded the drone flight during beam exposure. A microphone was also placed near the drone to record audio during the drone flight. Lastly, Wi-Fi connection data were captured using Wireshark [21].

IV. Results

Potential operational failures due to SEUs may result in a crash (drone flight becomes unstable), veer off the intended path, communication failure between laptop and drone, and/or sensor failure (incorrect data is sent to flight controller). The use of telemetry allows to identify deviation from normal operation by comparing the telemetry data of a flight run that experienced failure due to SEU with a flight run that was not exposed to the neutron beam (called the "golden copy"). Telemetry data shows various attributes of the flight that describe the drone's intended behavior. Thus, any behavior that deviates from the golden copy is assumed to be an anomaly and assumed to be a direct result of beam exposure.



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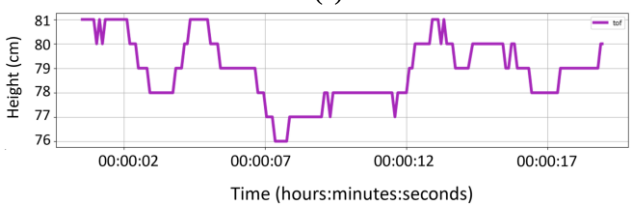
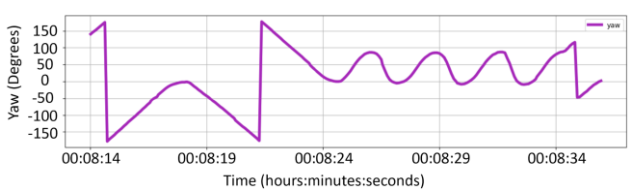


(b)

Fig. 2. Golden copy of telemetry data for (a) yaw and (b) height. The yaw signal regularly increases and decreases with a range of 180 degrees and -180 degrees for a full clockwise and counterclockwise movement.

Fig. 2 shows the golden copy of the telemetry data for yaw and height. The yaw should increase and decrease at regular intervals, indicating the clockwise and counterclockwise movement of the drone during the experiment. The height varies slightly during the run but should not have large fluctuations.



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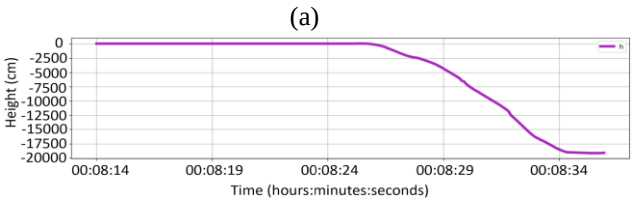


(b)

Fig. 3. Yaw data shows clear deviations from the golden copy in (a) while a few seconds later the height data shows clear deviations in (b) when the crash occurs. Note that both graphs show a small-time duration right before the crash and do not show the data of the entire run.

Fig. 3 shows the telemetry for yaw and height during one of the runs for which an operational failure was observed. Yaw data, as expected, varies similar to what is seen in the golden copy until the occurrence of an assumed SE upset. After which, the yaw data shows a clear deviation from the golden copy. Height data for the same test run shows a crash to the ground a short time later. The drone started losing altitude a few seconds after the SE upset affected yaw. The telemetry data in Fig. 3b shows impossibly low height values. The physical motion of the drone flipping upside down because of the SE-effect could cause the infrared sensor to not see the ground properly and lead to improper values for height. If sensor data was not affected by any physical motions, then most likely this data came from micro-controllers miscalculating the height data. It is possible that sensor data got corrupted on the way from the sensors to the CPU or from the CPU to the telemetry receiver, however either one of these errors would have been corrected immediately by the next stream of data from the sensor. It is most likely that the sensor is not able to see the ground due to the physical flipping of the drone or miscalculation due to drone firmware, and thus can be qualified as a transient failure.

In total, out of 28 runs, two of the runs did not experience any failure before the drone's battery ran out. Amongst the remaining runs, there were two types of failure modes observed, (i) early landing failures where the drone landed (Somewhat) gently, (ii) drone crash landed where the drone mostly dropped to the ground. For the runs when a failure occurred, failure characteristics, associated FIT rates, and the time gap between anomaly detection and actual operational failure are listed in Table I.

Failure Mode	**FIT Rate (failures/10^9 device hours)**	**Average Time Gap Between Anomaly and Failure (seconds)**
Crash	7	14.8
Erratic Flying	43	No time gap was observed
Early Landing	113	8.6
Overall	87	11.8

Table I. FIT rates for various failure modes show a wide range. While the overall FIT rate was 87, runs with an early landing failure characteristic had the highest FIT rate while crash had the lowest FIT rate.

For each run, parameters that showed anomalies first in the telemetry data stream were recorded and are listed in Table II. Overall, acceleration in the z direction was the parameter that most commonly experienced anomalies first. Similar to the explanation for Fig. 3b, it is likely that SE-induced crash events cause the drone to be flipped upside down, making infrared sensors not see the ground, and thus lead to anomalous values for multiple parameters.

Parameter that failed first	Number of times
Acceleration (z direction)	13
Time of Flight	11
Velocity (z direction)	10
Acceleration (x direction)	8
Height	6
Acceleration (y direction)	6
Yaw	5
Pitch	5
Roll	4
Velocity (y direction)	4
Velocity (x direction)	3
Barometer	1

Table II. Parameters that were recorded to show anomalies first across all the runs regardless of failure mode are listed in the table. Note that if multiple parameters failed at the same time, they were all counted as "first" failures.

These parameters that failed first were also then categorized by these failure modes (early-landing failures and crash failures). For runs with early-landing failures, time-of-flight was the parameter that was observed to fail first most often. For runs with crash failures, acceleration in the z direction was most commonly observed to fail first. It should be noted that this may be correlation more than causation as some failures will cause the drone to quickly lose altitude, which entails anomalies for acceleration in the z direction. Telemetry data may not always display the cause but show the effect of an unseen cause. The parameters that failed first based on specific failure modes can be seen in Table III. and Table. IV.

Parameter that failed first	Number of times
Time of Flight	7
Velocity (z direction)	6
Height	5
Acceleration (z direction)	5
Acceleration (x direction)	4
Acceleration (y direction)	3
Pitch	2
Yaw	1
Roll	1

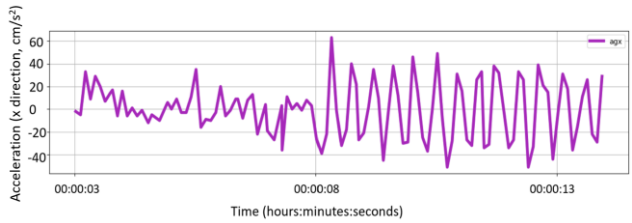
Table III. Parameters that were recorded to show anomalies first in runs with an early landing failure are listed in the table, from most common to least common. Again, note that if multiple parameters failed at the same time, they were all counted as "first" failures.

Parameter that failed first	Number of times
Acceleration (z direction)	8
Acceleration (x direction)	4
Yaw	4
Velocity (y direction)	4
Time of Flight	4
Velocity (z direction)	4
Acceleration (y direction)	3
Pitch	3
Roll	3
Velocity (x direction)	3
Barometer	1
Height	1

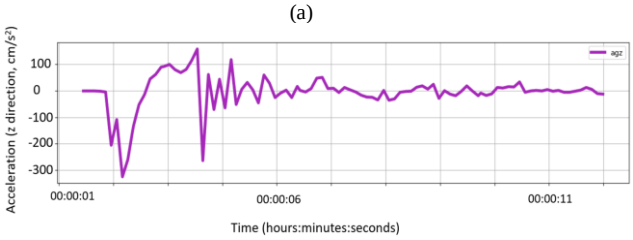
Table IV. Parameters that were recorded to show anomalies first in runs with a crash-landing failure are listed in the table, from most common to least common. Again, note that if multiple parameters failed at the same time, they were all counted as "first" failures.

For the failures observed during these tests, there was typically a lag time of a few seconds between the first anomaly in telemetry data and the physical failure of operation. The time duration between the anomaly in telemetry data and the actual crash varied. For the early-landing failures, there was a smaller time gap between telemetry data anomaly and operational failure. The time gap in runs with early-landing failures ranged from 0 seconds (anomaly and failures occurred within too short a time for measurements) to up to 50 seconds. Most early-landing failures showed a time gap between anomaly and operational failure between 3 to 5 seconds. Comparatively, the

time gap for crash failures ranged from 0 second to 70 seconds. Seeing no time gap between anomaly and operational failure in runs with crash failures (5 out of 12 trials) was slightly more common than early-landing failures (3 out of 10 trials). However, it should be noted that anomalies observed in runs with crash failures could last throughout the majority, or even sometimes the entirety, of the run and then telemetry data would suddenly cut off at the time of the crash (5 out of 12 trials), whereas this was not observed in runs with early-landing failures. Fig. 4 shows telemetry data for such an event where the anomaly in telemetry data continued for a (relatively) long time before the actual crash occurred. Such a behavior is highly dependent on drone firmware, and it is unwise to draw any conclusions regarding the fault location based on test data.



(a)



(b)

Fig. 4. Graphs in (a) show a run in which there was a crash failure. Acceleration in the x (top graph) and z (bottom graph) directions show anomalies appearing for approximately six seconds before the crash was observed. Note that the graph for acceleration in the z direction was shifted up approximately 1 g to remove the gravity component for ease of readability. The graph in (b) shows a run in which there was a land failure. There was a single, clear anomaly in the acceleration in the z direction 14 seconds before the failure was observed visually. Note that all three graphs show a specific time frame and do not show the data of

The difference in the length of the anomaly between runs with early-landing failures and crash failures may be attributed to Tello EDU drones' safety mechanism of a forced landing command, which is pre-built into the drone's firmware. When no inputs are sent to the drone (i.e., the drone simply hovers in the air with no additional input) or the drone suspects some error has occurred (e.g., Wi-Fi connection was compromised, battery level has depleted below 10%, etc.), the drone automatically lands itself as a safety mechanism to protect itself from potential damages. Thus, when only a few parameters fail or the failures are not extreme, the drone may take preemptive measures to land itself which shows as a time gap between telemetry anomaly and visual failure. Conversely, runs where the drone experienced crash failures may experience multiple parameters failing at once (as seen by more parameters observed to have anomalies "first" in Table II. compared to Table I.) or telemetry data anomalies throughout the run that amplify into larger, catastrophic failures. These catastrophic failures may also prevent telemetry data from being sent to the remote control, which then results in no time gap being observed between telemetry and visual failures. For runs where telemetry data shows anomalies for a sustained period of time, an SE-event may have affected certain parameters but not to a degree which causes the drone to crash or execute an automatic safety landing. However, another SE event at a later point in time could affect those parameters again to have more extreme anomalies, causing a crash event and an abrupt stop in telemetry data being sent. Due to the complexity of the system, it is possible that multiple bit upsets may be required before some of the failures are observed. In some cases, anomalies in multiple parameters may be needed (either simultaneously or accumulated over time) to cause a crash.

Such telemetry data indicates that though drone operators may not be able to react in time to avoid SE-induced failure, it may be possible to regain control of the drone with an automated correction feature before the forced landing or crash happens. Telemetry data from another run for pitch roll, and yaw parameters before the crash are shown in Fig. 5. Telemetry data shows large fluctuations in the yaw, pitch, and roll values a few seconds prior to a crash. With proper detection and correction within this time gap, it may be possible to detect the anomaly and have firmware-based recovery.

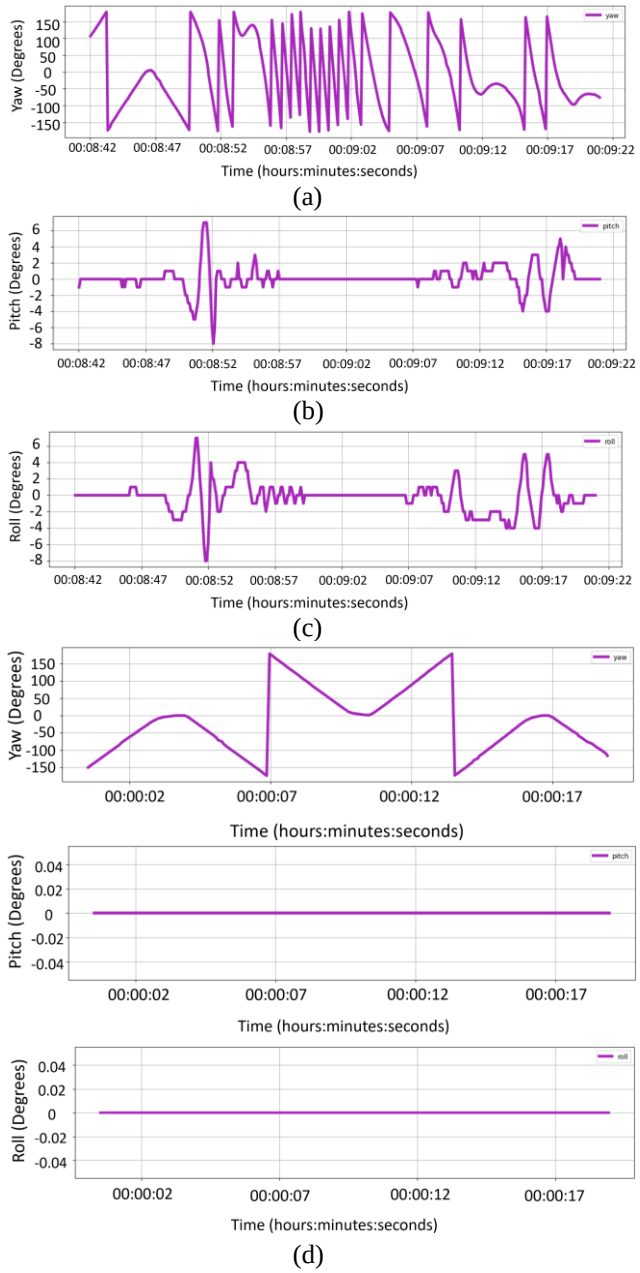


Fig. 5. Yaw (a), pitch (b), and roll (c) data all show clear deviations from the golden run a few seconds before the actual landing. (d) shows golden data for yaw, pitch, and roll, respectively. Note that all three graphs show a specific time frame and do not show the data of the entire run.

Failure in any one of the components in the drone will take time to go through the autonomous system. Additionally, failures will need to be in an area that is not constantly updated (for example an error in yaw data will be overwritten by the next sampling and most likely won't cause a failure). As a result, most errors in the sensor module and motor controller will cause anomalous behavior only for a short time (erratic flying type of failures in Table I) and will be eventually corrected by the flight controller. All drones that experienced a loss of function were fully operational after a power cycle. Since it can be reasonably inferred that all the telemetry is from monitor sensors, the drone malfunction is likely caused by a failure in code execution in the flight controller or in the component circuit that plays a key role in the drone's flight functions.

Drones from various companies may have different COTS hardware components, and a drone's software may be brand dependent as well. Additionally, as mentioned before, several components likely failed at the same time based on the telemetry data that was gathered. Thus, a system-level approach to mitigate SEEs may be necessary to devise a more general solution that can apply to drones even with different hardware components. Anomalies can likely be detected using software mitigation techniques, cycles are available between anomaly detection and actual failure, and the physical position and velocity of the drone itself may allow enough time for recovery. Especially with a pre-programmed flight, software could be written to detect when clearly erroneous instructions are being sent or incorrect data being recorded (as in Fig 3b. where an impossible -20,000 cm height was recorded). For transient sensor errors that affect the motor controller where the flight control firmware runs, it could be possible for a watchdog monitor of the motor controller to reset the motor controller (or other circuitry / software monitoring the drone) and stabilize the drone before a catastrophic failure. Designers could develop auto-safety features where these incorrect parameters are prevented from affecting drone operation.

V. Conclusion

Drones assembled with COTS were exposed to a terrestrial neutron beam. Results show high vulnerability to SE effects with FIT rates for different failure mechanisms in the 43 to 113 range for the tested drones. Telemetry data for many parameters associated with the drone were collected and analyzed to elucidate underlying failure mechanisms. Time gaps between an anomaly in telemetry data and the observed failure were found and may be used to correct SE-induced operational failures, even for applications outside of the drone industry. Systems that require various components to work simultaneously for successful operation, such as in drones, could use software to detect anomalies and take corrective action before an operational failure occurs.

Acknowledgment

The authors greatly appreciate the staff at the ISIS Neutron and Muon Source in the STFC Rutherford Appleton Laboratory for radiation experiments performed at ChipIR.

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